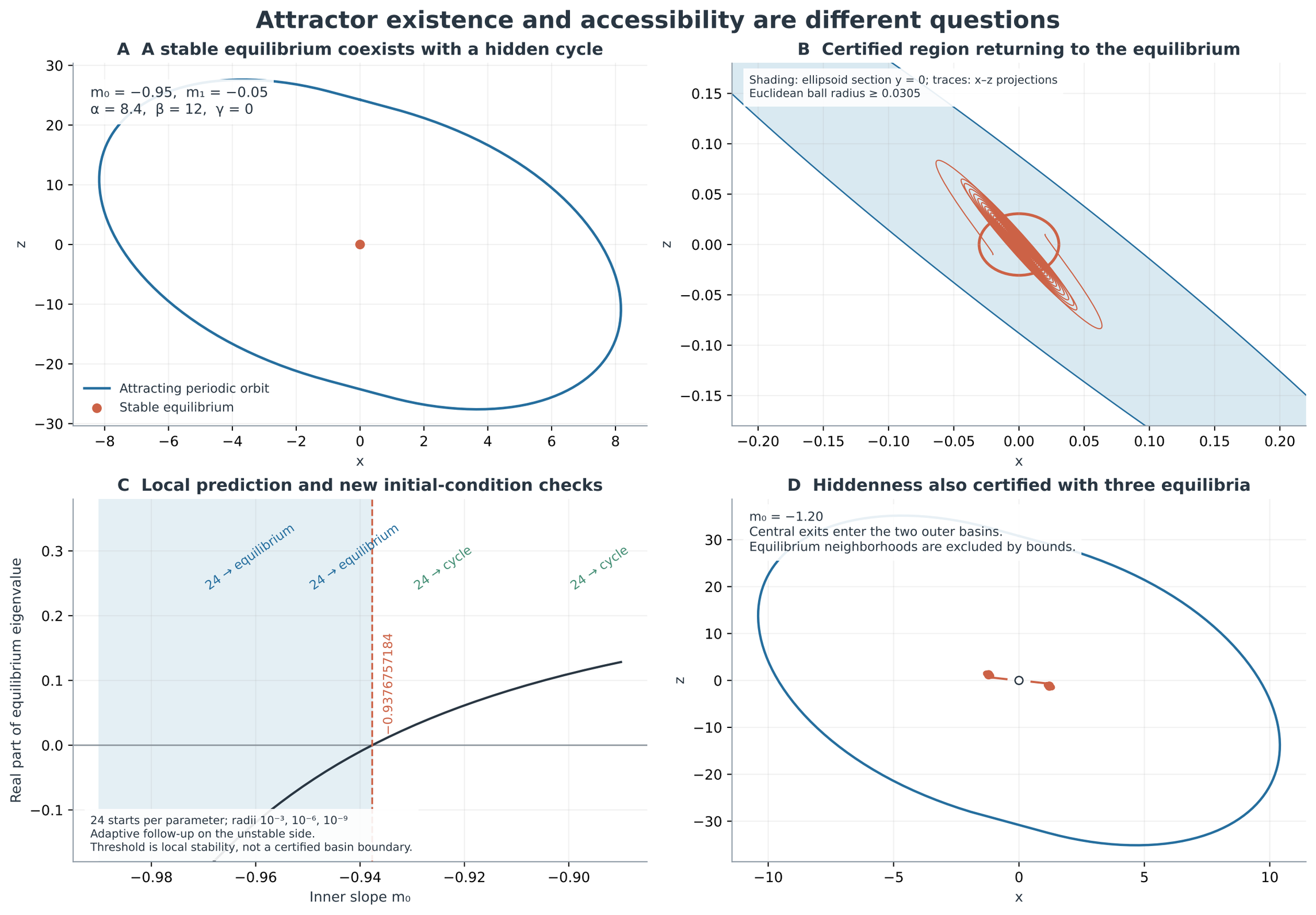

Attractor existence and accessibility are different questions
A A stable equilibrium coexists with a hidden cycle
m₀ = −0.95, m₁ = −0.05
α = 8.4, β = 12, γ = 0
Attracting periodic orbit
Stable equilibrium
x
z
B Certified region returning to the equilibrium
Shading: ellipsoid section y = 0; traces: x–z projections
Euclidean ball radius ≥ 0.0305
x
z
C Local prediction and new initial-condition checks
24 → equilibrium
24 → equilibrium
24 → cycle
24 → cycle
−0.9376757184
24 starts per parameter; radii 10⁻³, 10⁻⁶, 10⁻⁹
Adaptive follow-up on the unstable side.
Threshold is local stability, not a certified basin boundary.
Real part of equilibrium eigenvalue
Inner slope m₀
D Hiddenness also certified with three equilibria
m₀ = −1.20
Central exits enter the two outer basins.
Equilibrium neighborhoods are excluded by bounds.
x
z